

Fly.io

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: High

Key Metrics

Pages scanned: 1058

Report type: Premium Executive Audit

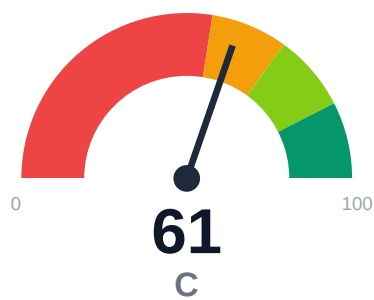
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 94
- Link verification inconclusive for 5 URLs (auth/rate-limit/server block); these are excluded
- SEO/GEO issue rate is high: 10.7%
- API usage-doc coverage is low: 60.0% (137 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Fly.io documentation spans 1,058 pages with 100% crawl coverage and strong retrieval readiness (82.11/100, 'strong' band). Critical issues include 94 confirmed broken internal links across user-facing docs, 60% API reference coverage (137 API pages detected), and 10.7% SEO/geo issue rate. Example reliability shows 0% detection across all pages, flagged as a likely JS-rendering detection limitation rather than true absence. Last-updated metadata covers only 55.86% of pages, creating staleness risk for 33.9% of content.

External posture: SEO/GEO issue rate **10.7%**, public API coverage **60.0%**, broken links **94**.

61 Audit Score	D Grade	High Risk Band
1058 Pages Scanned	94 Broken Links	10.7% SEO Issue Rate

Per-Site Breakdown

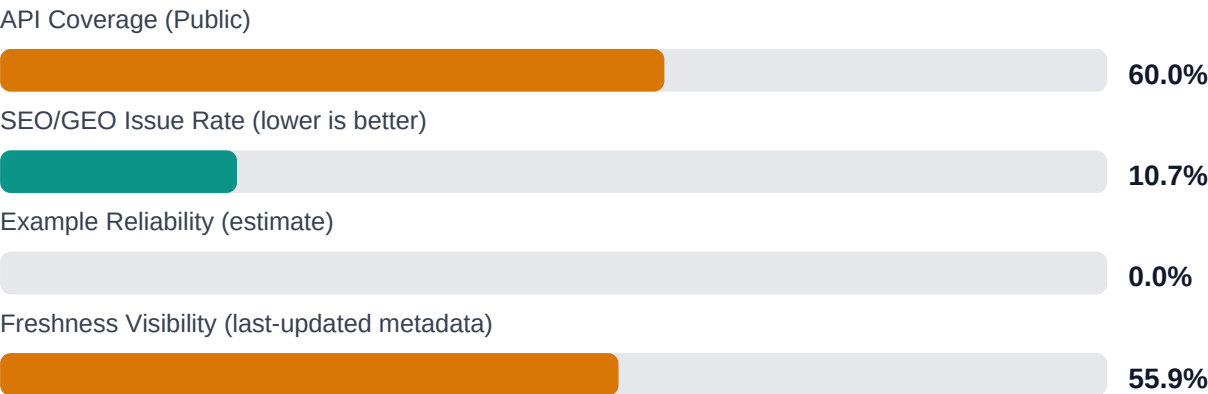
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://fly.io/docs	1058	94	60.0	0.0	10.7

Industry Benchmark Comparison

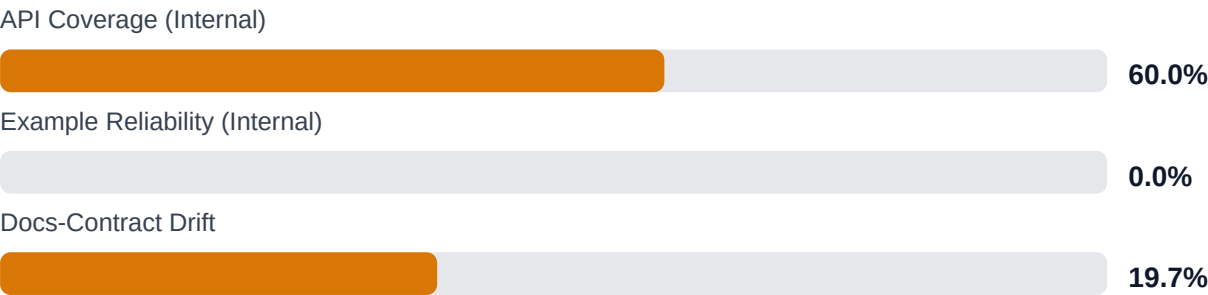
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-31
Industry average (developer docs)	85	-24
Minimum acceptable	70	-9
Your score	61	-

Gap to industry average: **24 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics

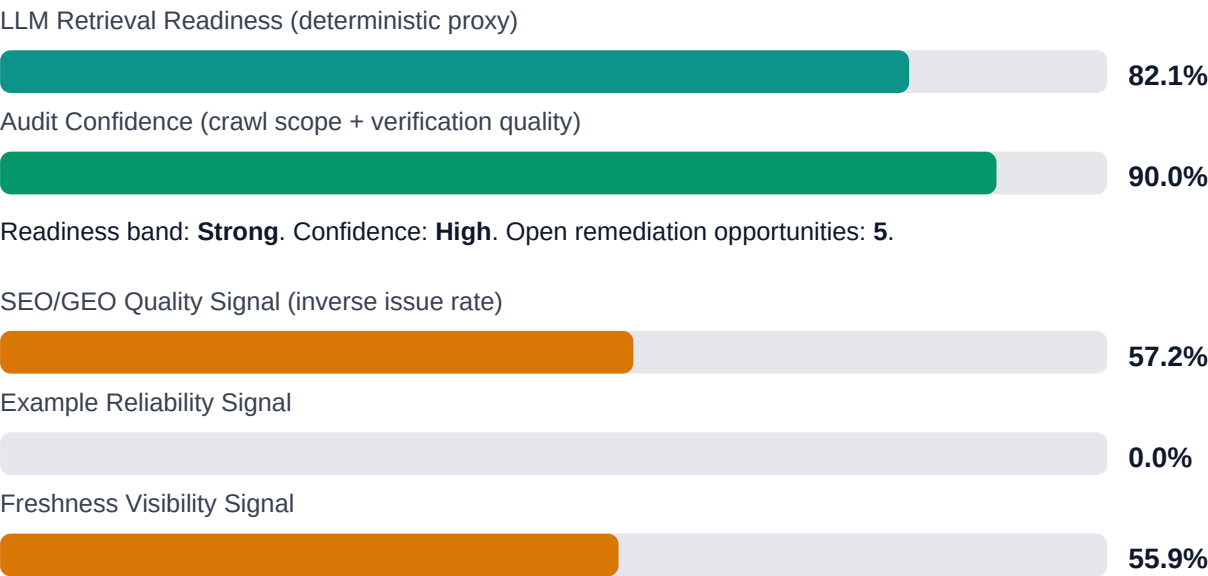


Internal Metrics (from scorecard)



Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.





LLM retrieval readiness breakdown (deterministic)



Evidence coverage: **80.3%** (1907/2376 key claims backed by deterministic evidence signals).

Highest-value remediation targets

Problem	Severity	Pipeline solution path
Internal documentation link integrity issues	HIGH	Weekly link governance via automated audits and consolidated remediation queue
Search and AI readability quality gaps	MEDIUM	Deterministic SEO/GEO linting and fix cycles before publish
Code example reliability risk	HIGH	Snippet lint + smoke checks + protocol self-verify coverage
Weak freshness and lifecycle visibility	MEDIUM	Lifecycle management, stale detection, and weekly governance cadence
API reference coverage and drift risk	MEDIUM	API-first contract flow, validators, regression gates, and drift checks

Financial Exposure Model

Low Estimate \$222,796 annual exposure Remediation: \$1,805 Monthly loss: \$3,562 Opportunity: \$14,853/mo	Base Estimate \$316,665 annual exposure Remediation: \$4,702 Monthly loss: \$11,144 Opportunity: \$14,853/mo	High Estimate \$474,033 annual exposure Remediation: \$7,600 Monthly loss: \$24,016 Opportunity: \$14,853/mo
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Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-API-COVERAGE	Undocumented API operations	HIGH	\$2,508	\$950
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH	\$1,938	\$808
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$969	\$475
F-DRIFT	Code/docs contract drift	MEDIUM	\$1,642	\$570
F-LAYERS	Missing required doc layers	LOW	\$1,539	\$712
F-TERMINOLOGY	Terminology inconsistency	LOW	\$598	\$332
F-RETRIEVAL	RAG retrieval quality risk	LOW	\$1,949	\$855

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-API-COVERAGE	Undocumented API operations	HIGH
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-DRIFT	Code/docs contract drift	MEDIUM
F-LAYERS	Missing required doc layers	LOW

Recommended Actions

1. Fix all 94 confirmed broken internal links across user-facing docs [impact: critical, effort: medium]
2. Audit and document the missing 40% of API surface (54 API endpoints or methods) to reach 100% reference coverage from current 60% [impact: high, effort: high]

3. Add or backfill last-updated timestamps for the 467 pages (44.14%) missing freshness metadata [impact: high, effort: low]
4. Resolve SEO/geo issues on approximately 113 pages (10.7% issue rate) to improve organic discoverability [impact: medium, effort: medium]
5. Investigate example detection failure (0% reliability) and either fix JS-rendering for audit tooling or manually verify top 50 high-traffic pages for working code examples [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage (1,058 of 1,059 discovered pages, 100%) with high crawl confidence (95%)
- + Strong retrieval readiness overall score (82.11/100) with 100% citationability and 89.51% answerability for AI/search use cases
- + Evidence coverage at 80.26% (1,907 of 2,376 key claims backed by evidence)
- + High chunkability (80.34%) supports effective content segmentation for RAG and search indexing
- + Actionability coverage at 100%, indicating clear next-steps guidance throughout docs

Risks

- 94 confirmed broken internal links degrade user navigation, trust, and SEO authority across docs
- API reference coverage at 60% leaves 40% of API surface undocumented or undetected, risking developer onboarding friction
- SEO/geo issue rate of 10.7% (approximately 113 pages) may suppress organic discovery and regional search performance
- Example reliability at 0% suggests detection failure; if examples are truly absent or non-functional, developer trust and time-to-first-success suffer
- Last-updated metadata missing on 44.14% of pages (467 pages) obscures content freshness and increases perceived staleness risk (33.9%)
- 5 unverified links due to auth/rate-limit/server blocks may hide additional broken links or access issues
- API confidence at 85% and example detection limitation reduce certainty in developer-experience metrics

Limitations

- * External audit cannot verify content technical accuracy, correctness of code examples, or alignment with actual platform behavior and API contracts.
- * Example reliability at 0% is flagged as a detection limitation due to likely JS-rendered code blocks; true example presence and functionality require manual or instrumented browser-based validation.
- * 5 unverified links blocked by auth, rate-limits, or server errors may hide additional broken links or access-control misconfigurations not surfaced in this scan.
- * API coverage confidence at 85% means detection may miss undocumented endpoints, especially if API pages lack consistent markup or are dynamically generated.
- * Audit does not measure user satisfaction, support ticket correlation, conversion impact, or whether docs answer the questions users actually ask.

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	95.0
support_hourly_usd	65.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	6.0
avg_release_delay_hours	18.0
monthly_support_tickets	420.0
docs_related_ticket_share	0.28
avg_ticket_handling_minutes	22.0
monthly_doc_influenced_evals	180.0
eval_to_customer_rate	0.12
avg_customer_monthly_value_usd	850.0
docs_friction_revenue_sensitivity	0.35

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	LOW

F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
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All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.